

Robustness Certification for Deep Neural Networks in Network Intrusion Detection Systems

Mengdie Huang, Elisa Bertino

December 8, 2025

<https://aiedge.osu.edu/sparks-students-and-postdocs-gathering-ai-research-and-knowledge-sharing>



Contents



Background



Problem



Solution

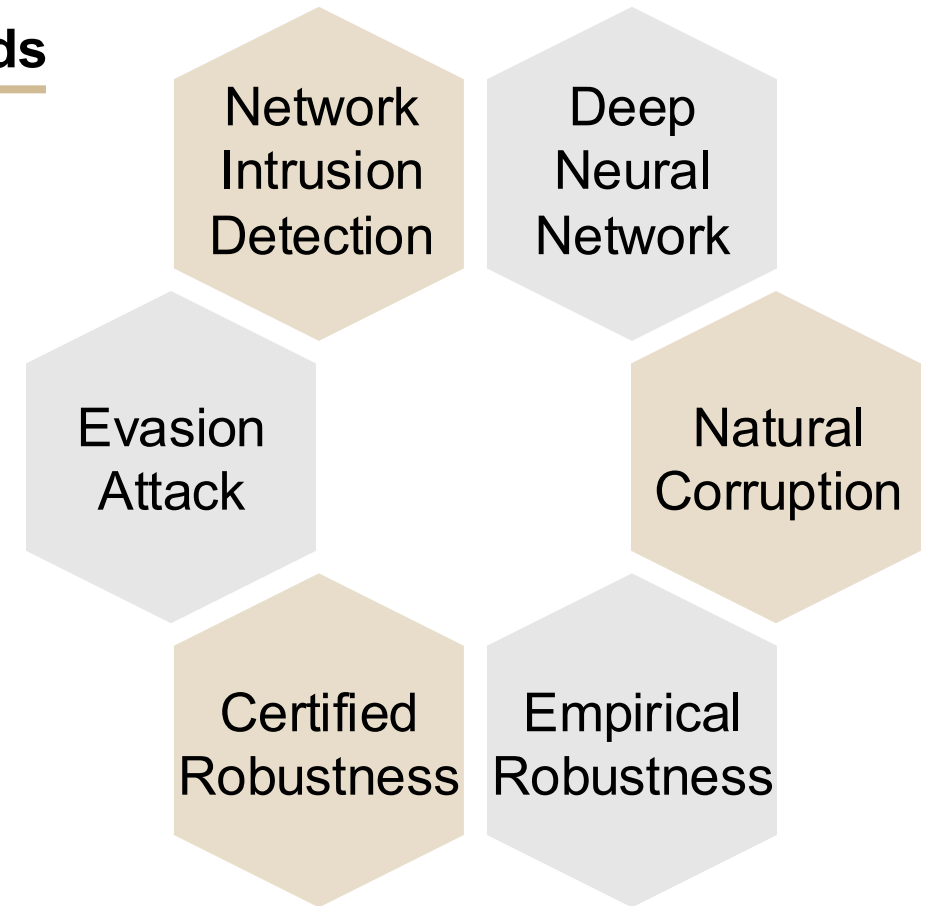


Evaluation



Conclusion

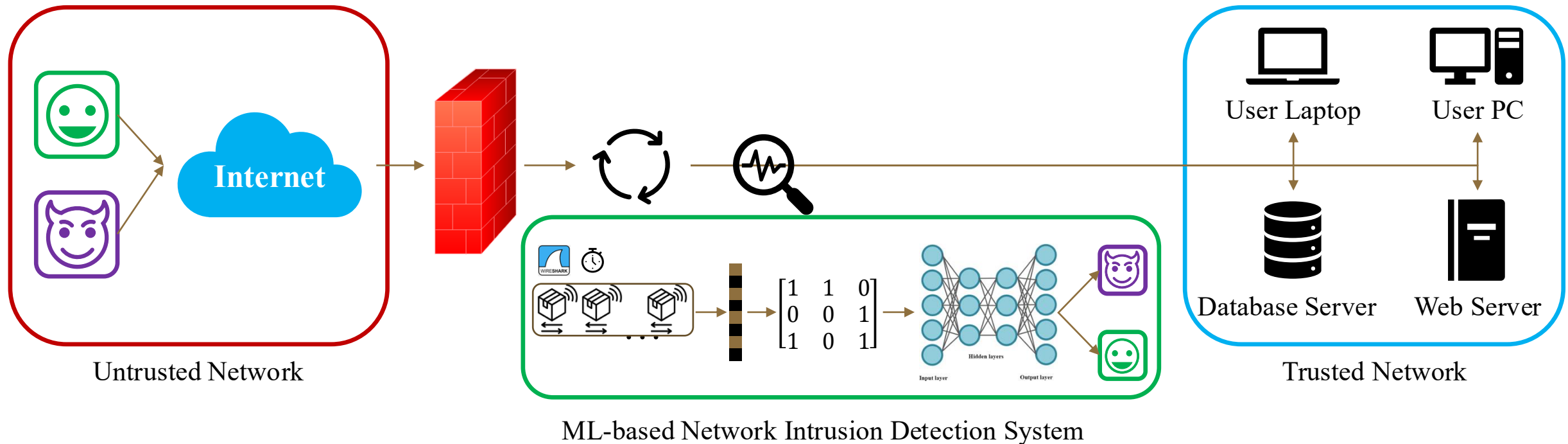
Keywords



Deep Neural Network-based Network Intrusion Detection System

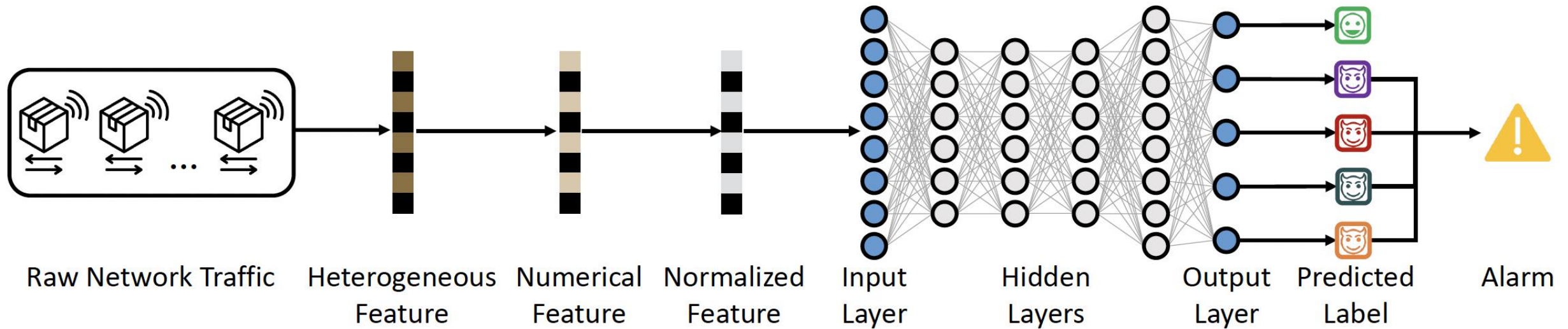
● Workflow of the DNN-based Network Intrusion Detection System

- A Network Intrusion Detection System (NIDS) is a security mechanism designed to monitor network traffic and identify malicious activities, such as unauthorized access, denial-of-service (DoS) attacks, probing, and malware propagation.
- A DNN-based NIDS leverages deep learning techniques to detect intrusions by learning discriminative patterns from large-scale network traffic data.



Deep Neural Network-based Network Traffic Classifier

- Workflow of the DNN-based Network Traffic Classifier



- Traffic Data includes both numeric and non-numeric values (e.g., protocol, network service, timestamp, etc.)
 - First, transform the raw network traffic vector x_{raw} into a numerical feature vector x_{num} .
 - Then, normalize it into a feature vector x in a continuous real number range.

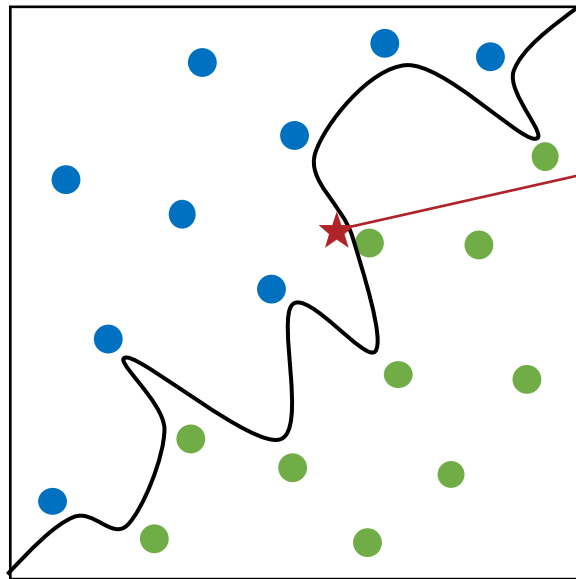
Threats to Deep Neural Networks (DNNs)

- Standard Train a Base Classifier F

- Optimization objective of standard training

$$\min_{\theta} \mathbb{E}_{(x, y_{true}) \sim \mathcal{D}_{train}} [\mathcal{L}(F_{\theta}(x), y_{true})]$$

Standard Training

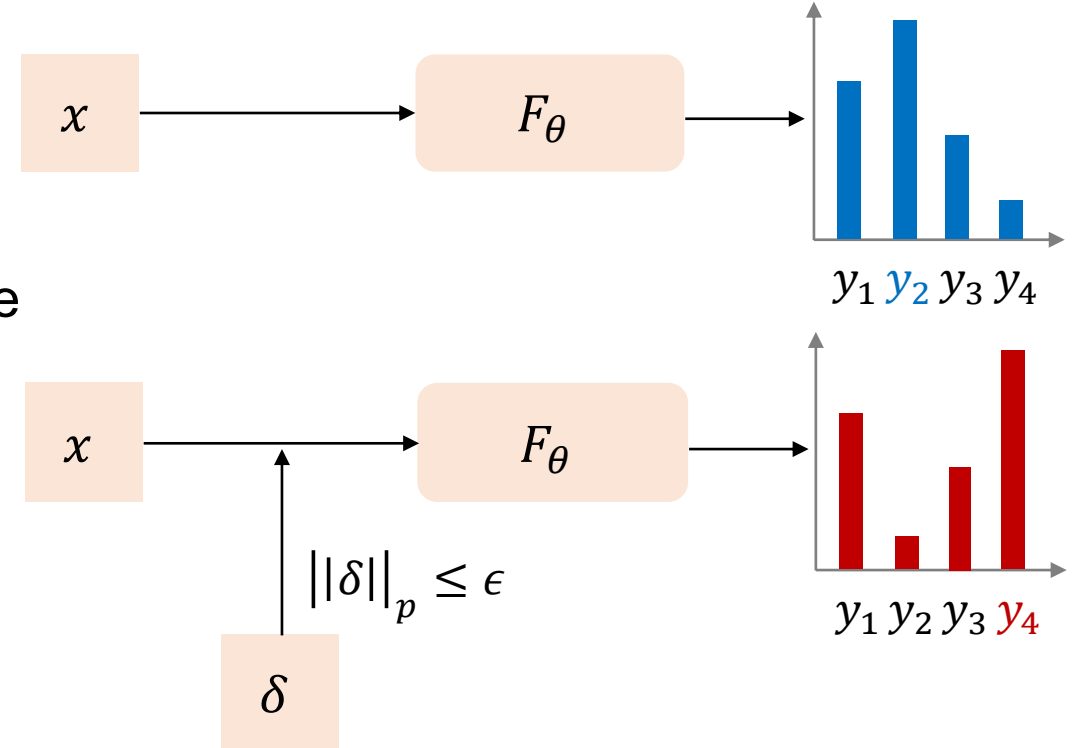


Adversarial Example

- Evasion Attack with Adversarial Example ($x + \delta$)

- Optimization objective of untargeted attack

$$\max_{\|\delta\|_p \leq \epsilon} \mathcal{L}(F_{\theta}(x + \delta), y_{true})$$



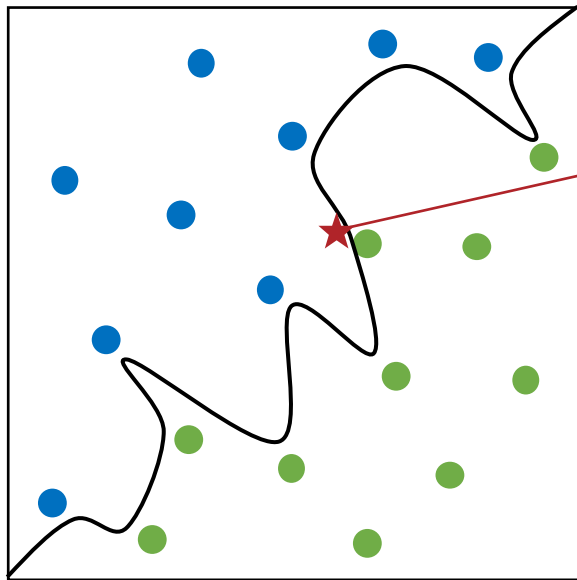
Empirical Defense vs. Certified Defense

- Perspective of Robust Defense for Deep Neural Networks (DNNs)

- Empirical Defense

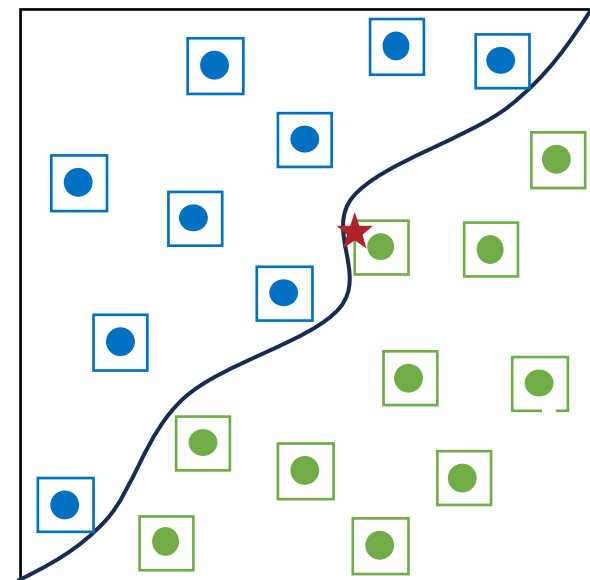
- Improve the model's prediction accuracy in adversarial attacks through robust training.

Standard Training



Adversarial Example

Adversarial Training



Empirical Defense vs. Certified Defense

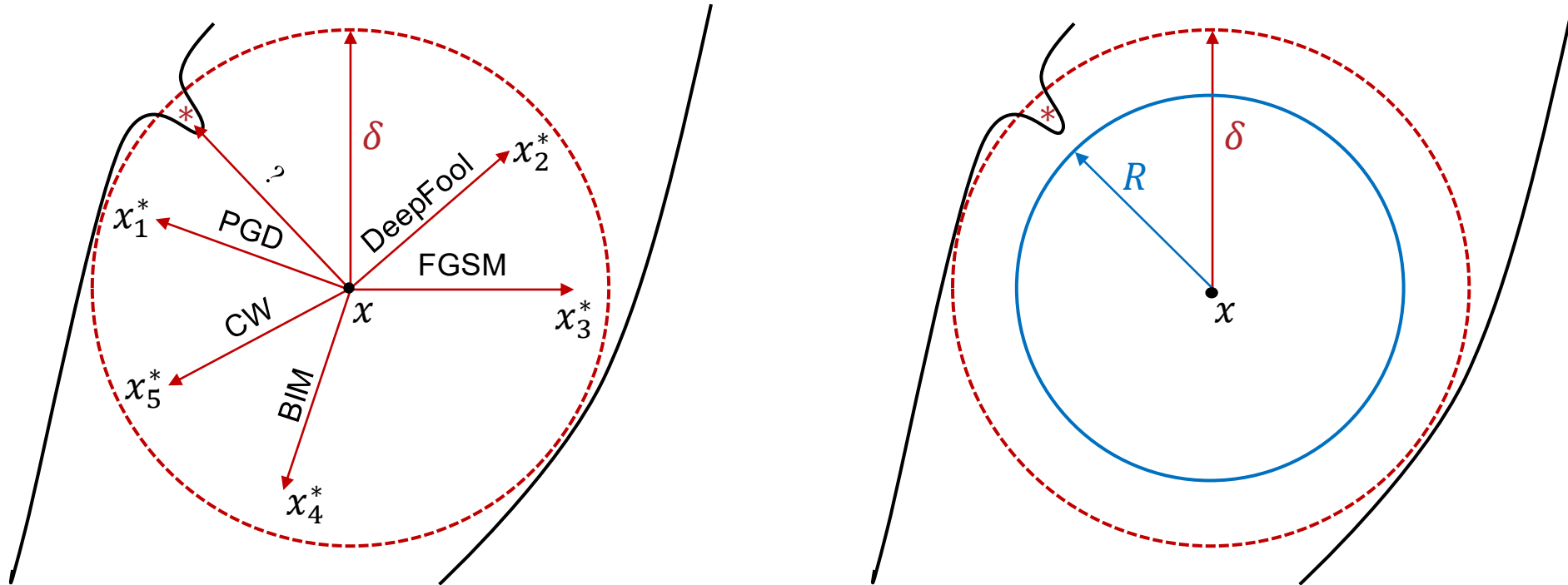
- Perspective of Robust Defense for Deep Neural Networks (DNNs)

- Empirical Defense

- Improve the model's prediction accuracy in adversarial attacks through robust training.

- Certified Defense

- Provide the certified robust radius R as the robustness certification of the predicted output.



Empirical Defense vs. Certified Defense

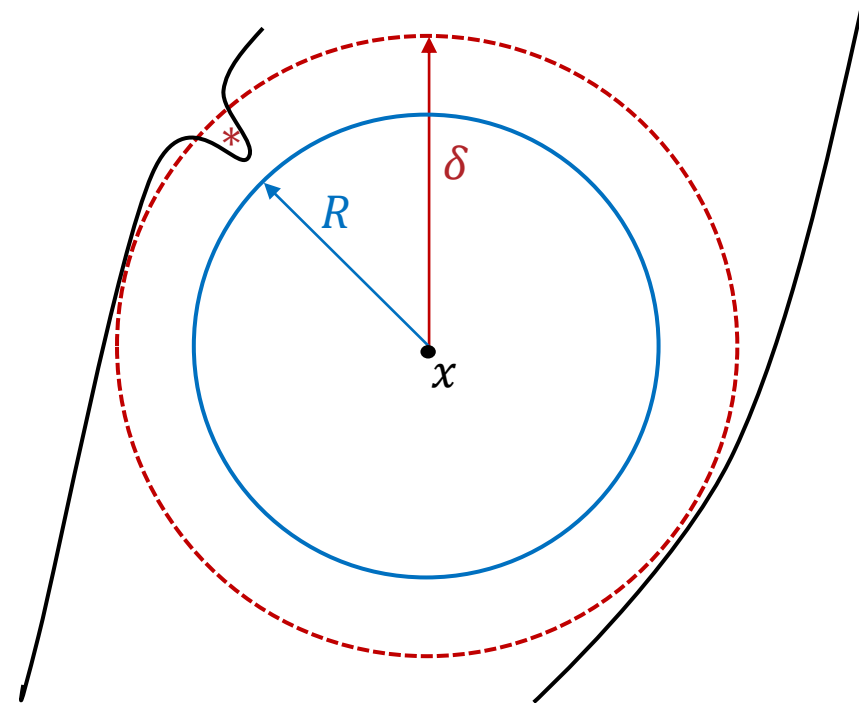
● Perspective of Robust Defense for Deep Neural Networks (DNNs)

➤ Empirical Defense

- Improve the model's prediction accuracy in adversarial attacks through robust training.

➤ Certified Defense

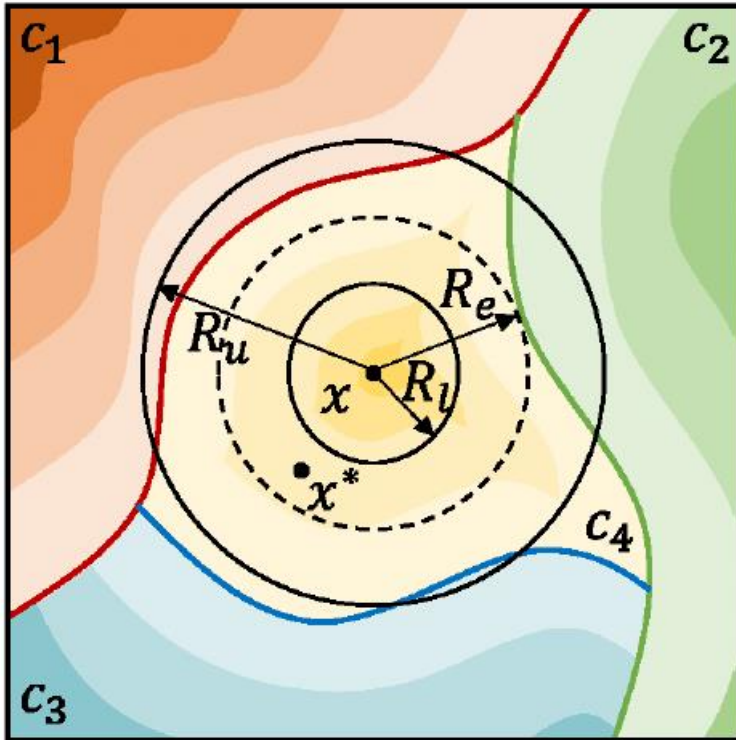
- Provide the certified robust radius R as the robustness certification of the predicted output.
- Robustness Guarantee
 - ✓ For input x , predictions of classifier F on perturbed data within an l_p norm-measured radius R around x , are guaranteed to remain consistent.
 - ✓ That is, any small perturbation δ to x within this region, including adversarial attacks, will not change the prediction results.



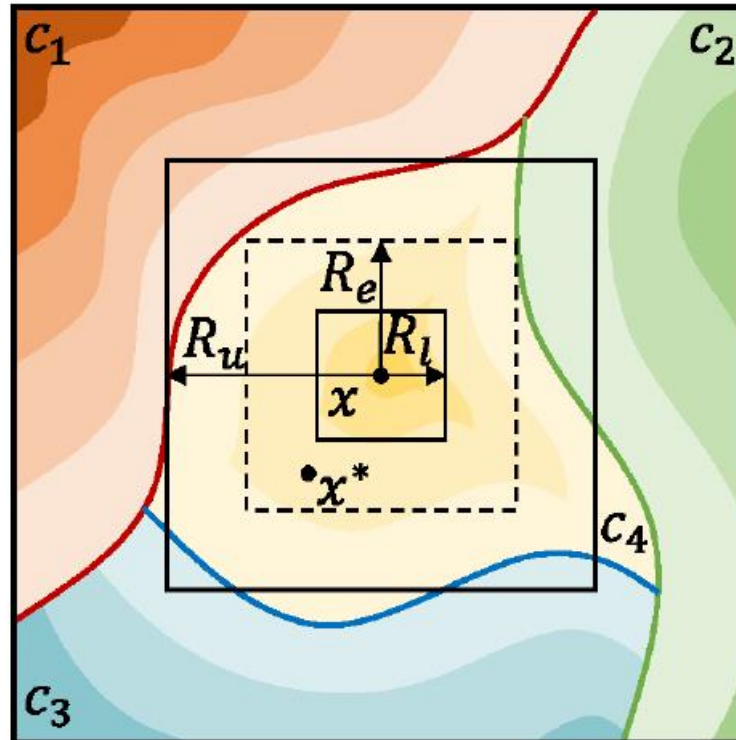
Certified Defense

- l_p Norm-bounded Certified Radius of DNN-based Multi-class Classifier on the Input x

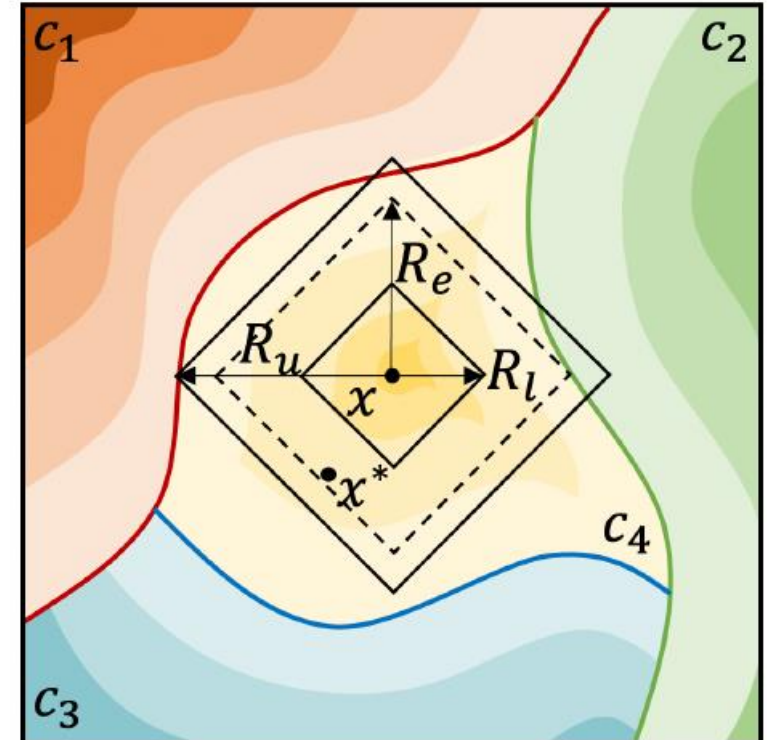
- Multiple Norm Types: l_2 norm, l_∞ norm, l_1 norm,
- Exact Robust Radius: R_e
- Upper/ Lower Bound of Exact Robust Radius: R_u, R_l



(a) l_2 radius: $\|\delta\|_2 < R$



(b) l_∞ radius: $\|\delta\|_\infty < R$



(c) l_1 radius: $\|\delta\|_1 < R$

Robustness Certification for DNN-Based Network Traffic Classifiers via MARS

- Robustness certification via randomized smoothing

DEFINITION 1 (BASE CLASSIFIER). Given a DNN model f_θ that maps input samples x to a C -dimensional confidence score vector, a base classifier F is defined as:

$$F(x) = \arg \max_{i \in [C]} f_\theta(x)_i,$$

where $f_\theta(x)_i$ represents the confidence score of the model f_θ on the i -th category for a given input x .

DEFINITION 2 (l_p ROBUSTNESS GUARANTEE). Given a classifier F , an input x , a perturbation δ , a distance measure l_p , robustness guarantee of F on x against l_p -bounded δ is defined as:

$$\text{For all } \delta \text{ such that } \|\delta\|_p < R, F(x + \delta) = F(x),$$

where R is the certified radius.

DEFINITION 3 (SMOOTHED CLASSIFIER). Given a base classifier F , an input x , a smoothing distribution $\mathcal{D}$ with Probability Density Function (PDF) φ , smoothed classifier F_{smooth} is defined as:

$$F_{smooth}(x) = \arg \max_{c \in [C]} \mathbb{P}_{\eta \sim \mathcal{D}}(F(x + \eta) = c) = \arg \max_{c \in [C]} \int_{\eta \sim \mathcal{D}} \mathbb{I}[F(x + \eta) = c] \varphi(\eta) d\eta,$$

where $\mathbb{P}$ is the probability function, $\mathbb{I}$ is the indicator function that returns 1 when the condition is true, and $\varphi(\eta)$ denotes the probability density of the sampled noise data η .

Certify Robustness of DNN-based Network Traffic Classifiers

- Motivation

- Certified defense efforts for network intrusion detection have been minimal, only BARS (NDSS'23).
- The l_2 robustness guarantee is relatively loose and lacks certification for other l_p certified radii.

- Problems to be solved:

- Pro1: Define a certified radius that can bound heterogeneous network traffic features.
- Pro2: Expand the certified robust region to tighten the robustness guarantee.
- Pro3: Provide the multiple l_p norms-bounded robustness guarantees of the model.

- Core Idea:

- Extend the real-value certified radius R to a vector $(R_1, \dots, R_d) \in \mathbb{R}^d$, where R_i denotes the dimensional certified radius for the i -th feature x_i of the heterogeneous input x .
- Introduce the multiple order information of the smoothed classifier to expand the certified region.
- Align the sampling area of smoothing distribution with the l_p -measured surroundings of the input.

Robustness Certification for DNN-Based Network Traffic Classifiers via MARS

- Robustness certification via randomized smoothing

DEFINITION 4 (ZERO-ORDER INFORMATION). Given a smoothed classifier F_{smooth} , an input x , the zero-order information refers to the basic characteristics of F_{smooth} , such as the statistical probability P_c that F_{smooth} predicts x as each class c in $[C]$, defined as:

$$\text{For all } c \in [C], P_c = \mathbb{P}(F_{smooth}(x) = c) = \mathbb{P}_{\eta \sim \mathcal{D}}(F(x + \eta) = c).$$

DEFINITION 5 (FIRST-ORDER INFORMATION). Given a smoothed classifier F_{smooth} , an input sample x , first-order information involves the derivative or gradient of F_{smooth} , defined as:

$$\begin{aligned} \text{For all } c \in [C], \|\nabla F_{smooth}^c(x)\|_p &= \|\nabla \mathbb{P}(F_{smooth}(x) = c)\|_p \\ &= \|\nabla \mathbb{P}_{\eta \sim \mathcal{D}}(F(x + \eta) = c)\|_p = \left\| \frac{\partial(\mathbb{P}_{\eta \sim \mathcal{D}}(F(x + \eta) = c))}{\partial(x)} \right\|_p, \end{aligned}$$

where $F_{smooth}^c(x) = \mathbb{P}(F_{smooth}(x) = c)$ is the statistical probability that F_{smooth} predicts x as c , and $\|\nabla F_{smooth}^c(x)\|_p$ is the l_p -measured magnitude of the gradient of F_{smooth}^c at x .

Certify Robustness of DNN-based Network Traffic Classifiers

- Motivation

- Certified defense efforts for network intrusion detection have been minimal, only BARS (NDSS'23).
- The l_2 robustness guarantee is relatively loose and lacks certification for other l_p certified radii.

- Problems to be solved:

- Pro1: Define a certified radius that can bound heterogeneous network traffic features.
- Pro2: Expand the certified robust region to tighten the robustness guarantee.
- Pro3: Provide the multiple l_p norms-bounded robustness guarantees of the model.

- Core Idea:

- Extend the real-value certified radius R to a vector $(R_1, \dots, R_d) \in \mathbb{R}^d$, where R_i denotes the dimensional certified radius for the i -th feature x_i of the heterogeneous input x .
- Introduce the multiple order information of the smoothed classifier to expand the certified region.
- Align the sampling area of smoothing distribution with the l_p -measured surroundings of the input.

Robustness Certification for DNN-Based Network Traffic Classifiers via MARS

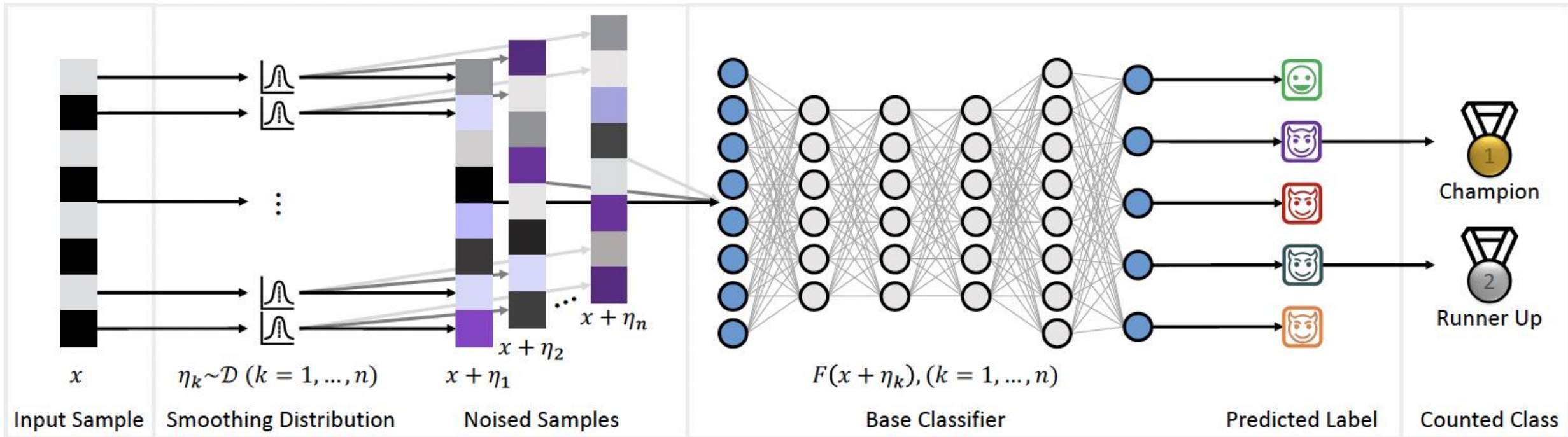
• Framework of Proposed Multi-Order Adaptive Randomized Smoothing (MARS)

➤ Prediction Procedure

- Sampling $n_k = n_{small}$ noise data $\rightarrow$ Predict the category of the input x .

➤ Certification Procedure

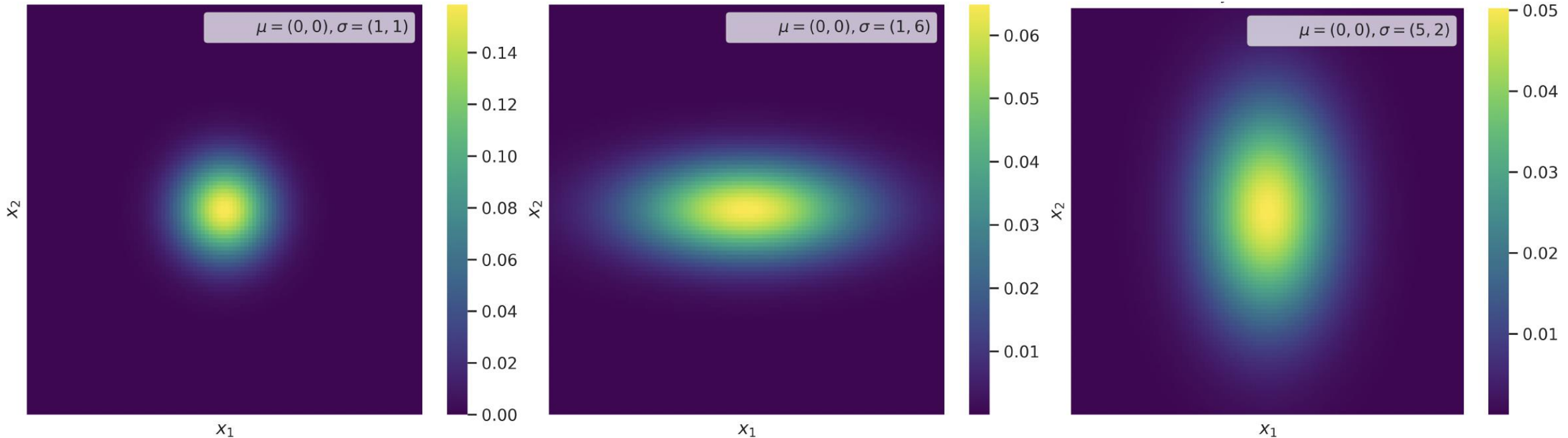
- Sampling $n_k = n_{large}$ noise data $\rightarrow$ Calculate the robust radius R of the model on the input x .



Robustness Certification for DNN-Based Network Traffic Classifiers via MARS

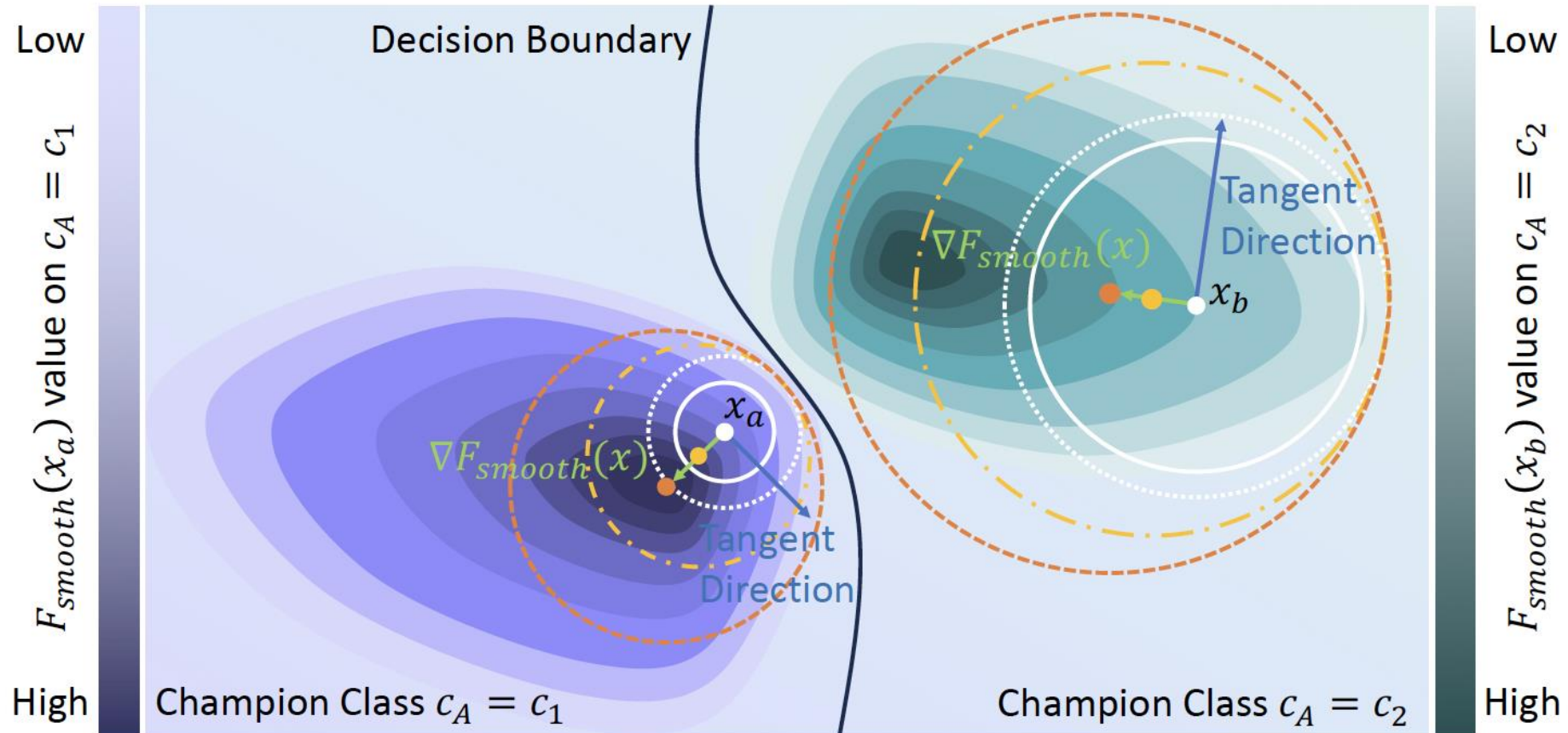
● Phase 1: Smoothing Distribution Parameters Optimization

- Distribution Shape Optimization.
 - Encourage noised samples to be near the decision boundary of the classifier for x .
- Distribution Scale Optimization.
 - Expand the noise sampling area by adjusting the smoothing distribution's scalar parameter.

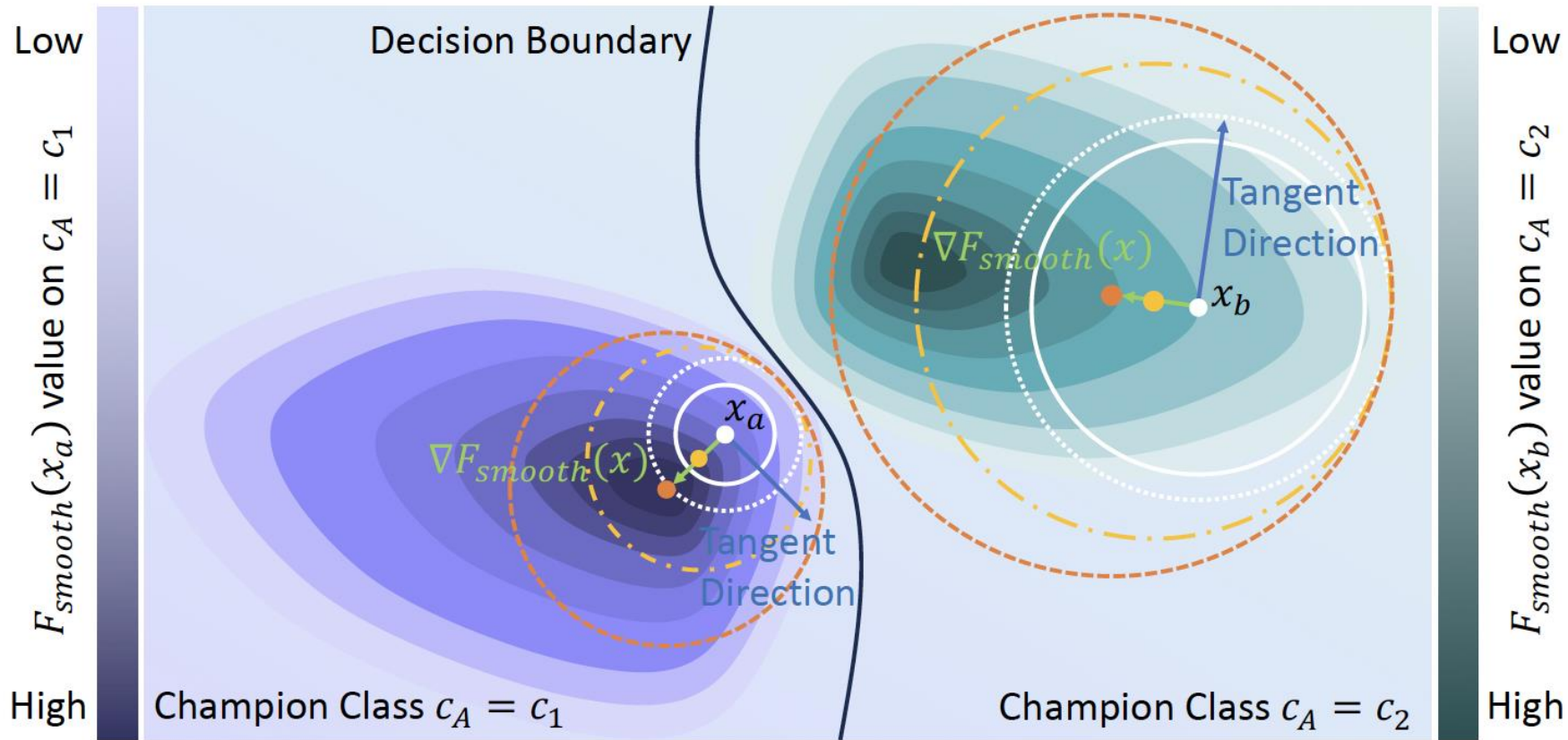


Robustness Certification for DNN-Based Network Traffic Classifiers via MARS

- Phase 2: Multi-order Information-based Certified Robust Radius Calculation
 - Zero-order Output Probability Information-based Certified Radius Calculation
 - First-order Gradient Information-based Certified Radius Extension



Robustness Certification for DNN-Based Network Traffic Classifiers via MARS



- c_A denotes the class with the highest confidence returned by the smoothed classifier F_{smooth} on the input.
- x_a and x_b are two inputs predicted as different champion classes, c_1 and c_2 .
- The solid white line outlines the certified region relying solely on zero-order output $F_{smooth}^{c_A}(x)$.
- The blue arrow indicates the tangent direction of $F_{smooth}^{c_A}(x)$ at x .
- The green arrow represents the gradient vector $\nabla F_{smooth}^{c_A}(x)$.
- The dotted white line outlines the certified region utilizing zero-order and a gradient with zero magnitude.
- The yellow and orange dotted lines outline the certified regions employing zero-order and a gradient with non-zero magnitude, where orange corresponds to a larger gradient magnitude.

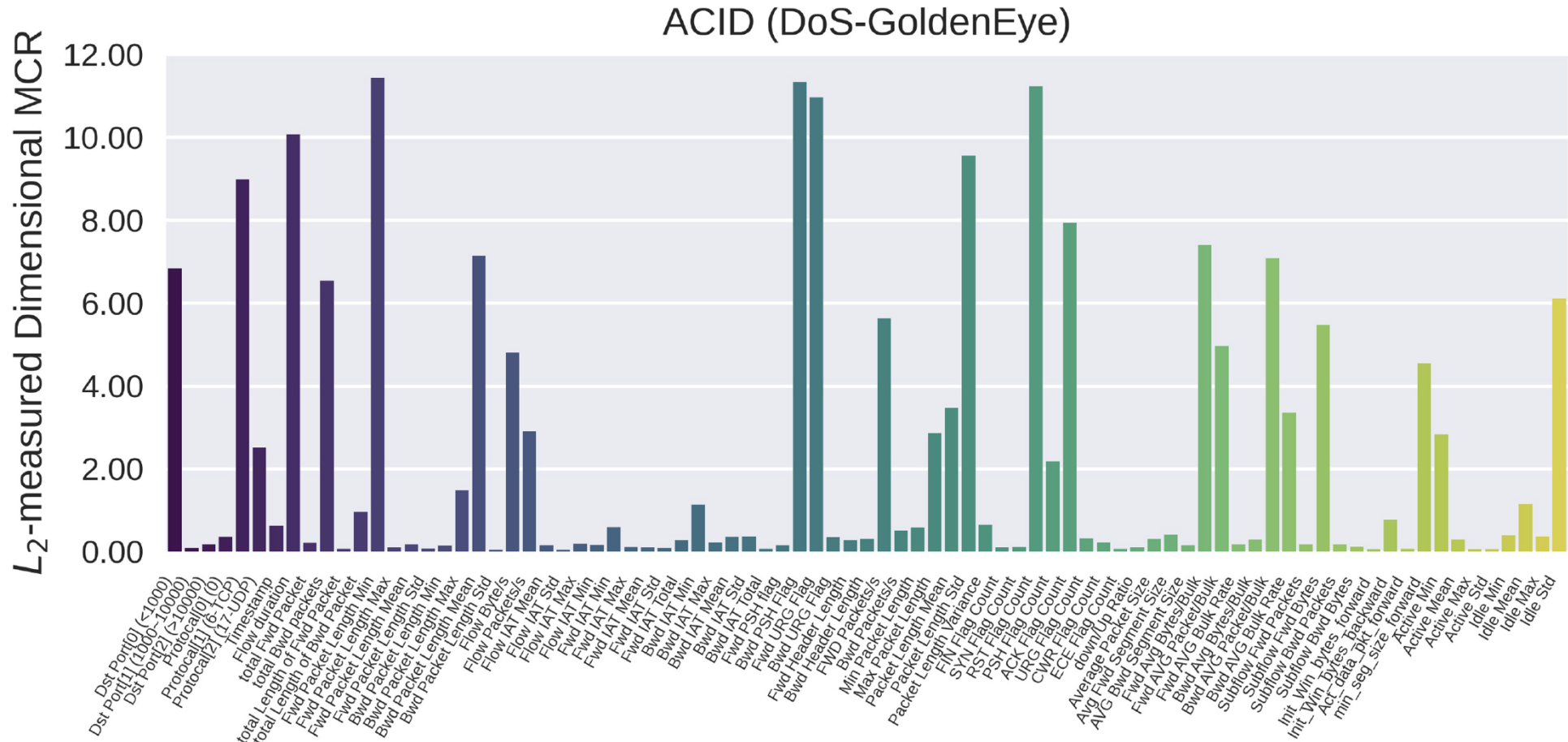
Robustness Certification for DNN-Based Network Traffic Classifiers via MARS

● Phase 3: Dimensional Robust Radius Weight Calculation Calculation

- Dimensional Feature Sensitivity Analysis
- Dimensional Radius Contribution Quantification

$$s_i = d(f_{\theta}^c(x))/d(x_i) \quad s = (s_1, \dots, s_d)$$

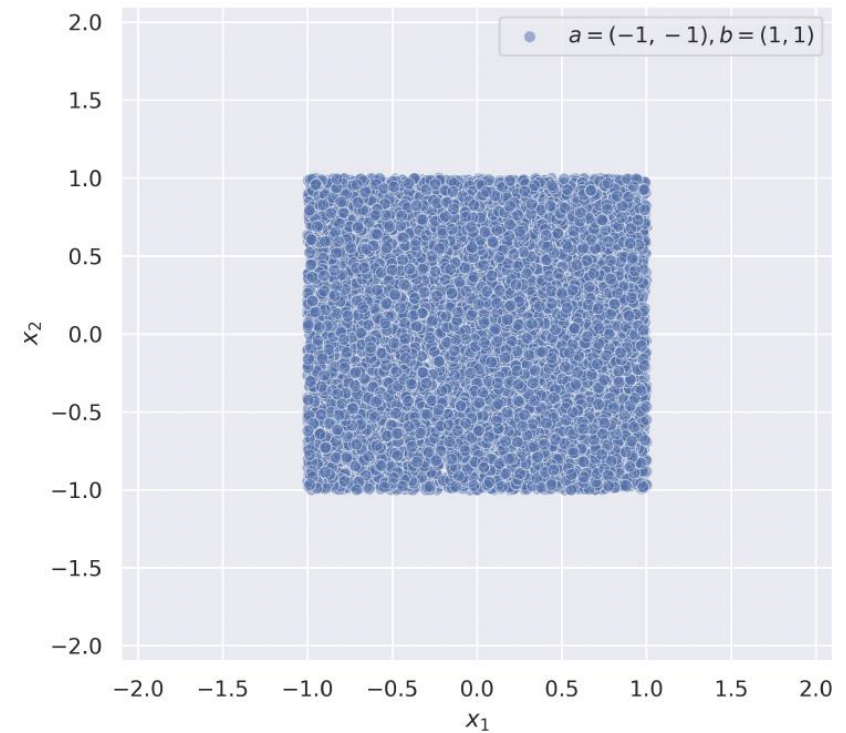
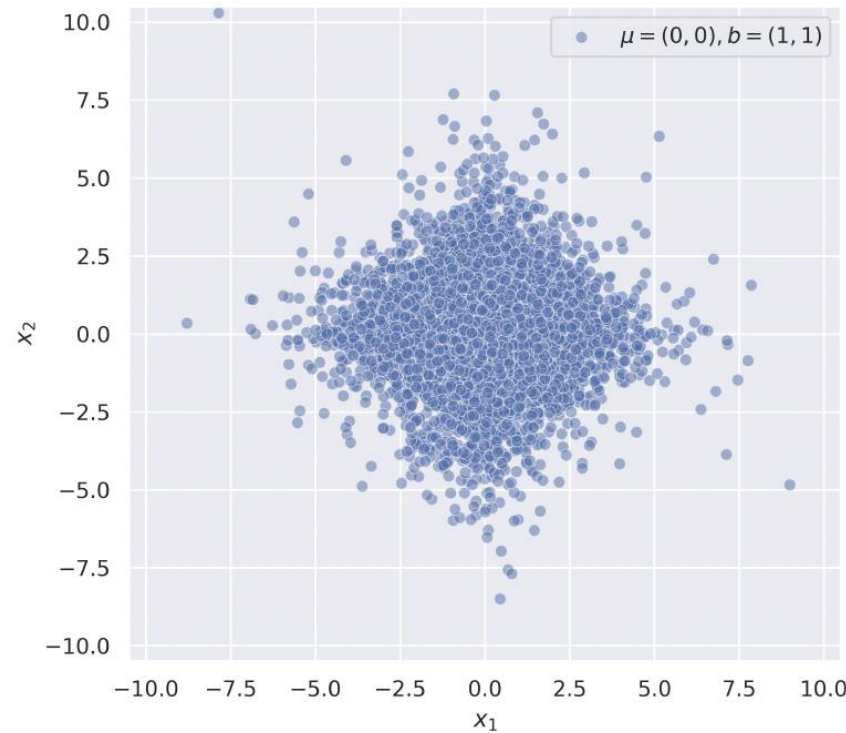
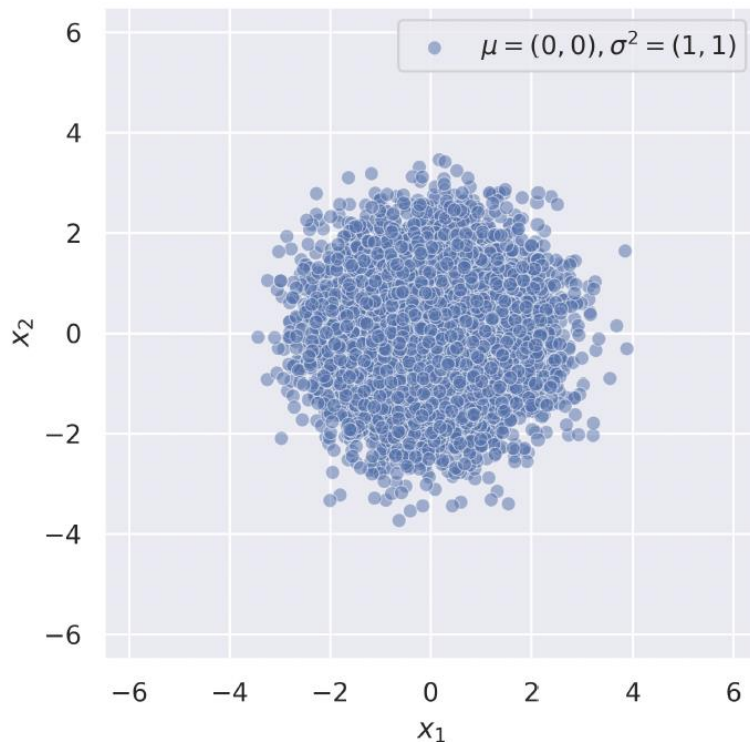
$$R_i = w_i \times R, w_i = \frac{R_i}{R} = \frac{1/d}{\tilde{s}_i} = \frac{1}{d\tilde{s}_i}$$



Robustness Certification for DNN-Based Network Traffic Classifiers via MARS

● Smoothing Distribution Diversity

- Gaussian Distribution aligns with l_2 norm-bounded certified region
- Laplacian Distribution aligns with l_1 norm-bounded certified region
- Uniform Distribution aligns with l_∞ norm-bounded certified region



Experiments

- Dataset

- Three datasets created from CIC-IDS-2018

- Model

- CADE: Contrastive Autoencoder for Drifting detection and Explanation (USENIX 2021)
- ACID: Adaptive Clustering-based Intrusion Detection (INFOCOM 2021)

Dataset	CSE-CIC-IDS-2018-CADE				CSE-CIC-IDS-2018-ACID			
	DoS-Hulk-Drift Dataset		Infiltration-Drift Dataset		Diverse-Intrusions Dataset		Similar-Intrusions Dataset	
	Class	Number	Class	Number	Class	Number	Class	Number
Training	Benign	52996	Benign	52996	Benign	52996	Benign	52996
	SSH-Bruteforce	9385	SSH-Bruteforce	9385	FTP-Bruteforce	12590	DoS-GoldenEye	26565
	Infiltration	7390	DoS-Hulk	34789	DDoS-HOIC	53476	DoS-SlowHTTPTest	11191
	-	-	-	-	Bot	22584	DDoS-LOIC-HTTP	46095
Test	Benign	13249	Benign	13249	Benign	13249	Benign	13249
	SSH-Bruteforce	2346	SSH-Bruteforce	2346	FTP-Bruteforce	3148	DoS-GoldenEye	6641
	Infiltration	1894	DoS-Hulk	8697	DDoS-HOIC	13369	DoS-SlowHTTPTest	2798
	DoS-Hulk	43486	Infiltration	9327	Bot	5646	DDoS-LOIC-HTTP	11524

Experiments

● Attack Configuration

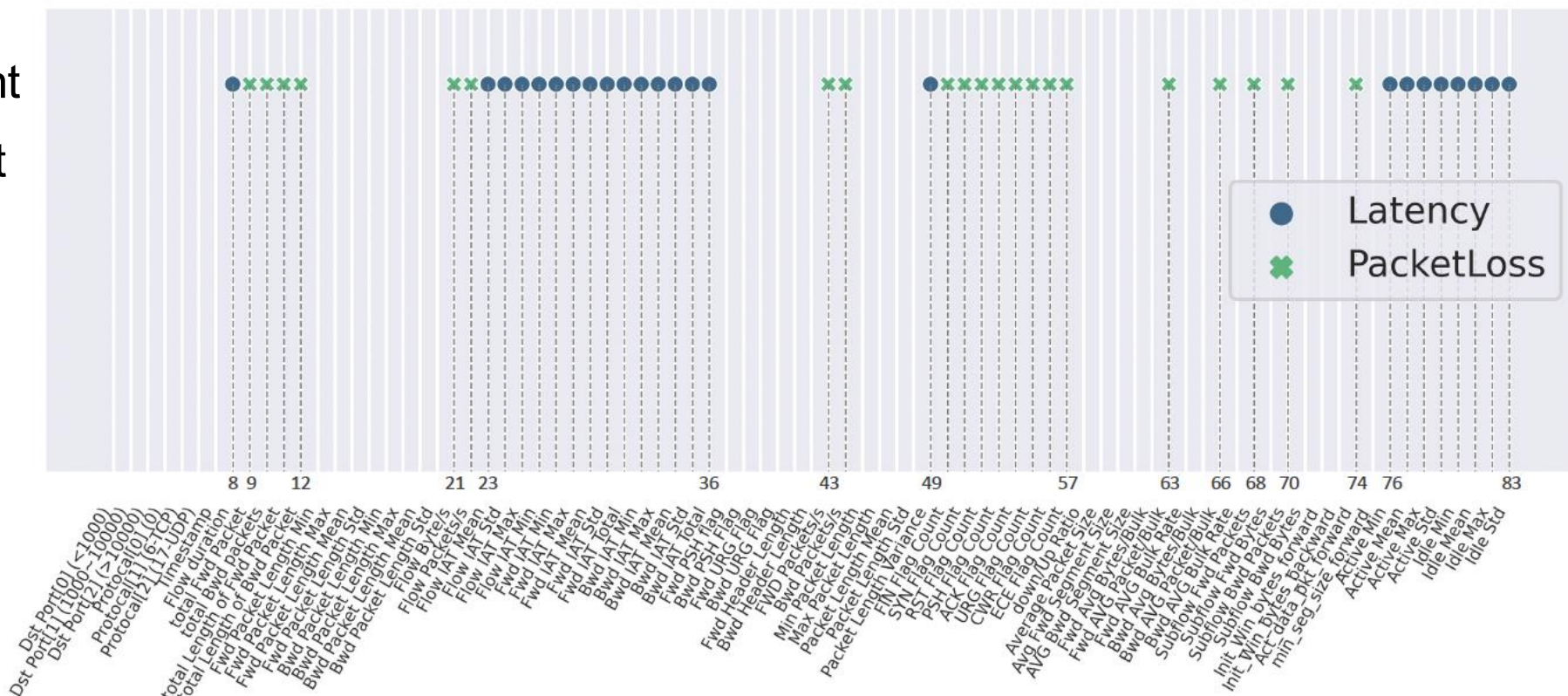
➤ Evasion Attack

- PGD: Projected Gradient Descent
- EAD: Elastic-Net Attack to DNN

➤ Natural Corruption

- Latency
- Packet Loss

Features Perturbed under Different Natural Corruptions



Experiments

● Attack Configuration

➤ Evasion Attack

- PGD: Projected Gradient Descent
- EAD: Elastic-Net Attack to DNN

➤ Natural Corruption

- Latency
- Packet Loss

Perturbed Featured under Latency

No	Feature Name	No	Feature Name
8	<i>Flow_Duration</i>	34	<i>Bwd_IAT_Mean</i>
23	<i>Flow_IAT_Mean</i>	35	<i>Bwd_IAT_Std</i>
24	<i>Flow_IAT_Std</i>	36	<i>Bwd_IAT_Total</i>
25	<i>Flow_IAT_Max</i>	50	<i>Packet_Length_Variance</i>
26	<i>Flow_IAT_Min</i>	76	<i>Active_Min</i>
27	<i>Fwd_IAT_Min</i>	77	<i>Active_Mean</i>
28	<i>Fwd_IAT_Max</i>	78	<i>Active_Max</i>
29	<i>Fwd_IAT_Mean</i>	79	<i>Active_Std</i>
30	<i>Fwd_IAT_Std</i>	80	<i>Idle_Min</i>
31	<i>Fwd_IAT_Total</i>	81	<i>Idle_Mean</i>
32	<i>Bwd_IAT_Min</i>	82	<i>Idle_Max</i>
33	<i>Bwd_IAT_Max</i>	83	<i>Idle_Std</i>

Experiments

● Attack Configuration

➤ Evasion Attack

- PGD: Projected Gradient Descent
- EAD: Elastic-Net Attack to DNN

➤ Natural Corruption

- Latency
- Packet Loss

Perturbed Featured under Packet Loss

No	Feature Name	No	Feature Name
9	<i>Total_Fwd_Packet</i>	53	<i>PSH_Flag_Count</i>
10	<i>Total_Bwd_packets</i>	54	<i>ACK_Flag_Count</i>
11	<i>Total_Length_of_Fwd_Packet</i>	55	<i>URG_Flag_Count</i>
12	<i>Total_Length_of_Bwd_Packet</i>	56	<i>CWR_Flag_Count</i>
21	<i>Flow_Byte/s</i>	57	<i>ECE_Flag_Count</i>
22	<i>Flow_Packets/s</i>	63	<i>Fwd_AVG_Packet/Bulk</i>
43	<i>FWD_Packets/s</i>	66	<i>Bwd_AVG_Packet/Bulk</i>
44	<i>Bwd_Packets/s</i>	68	<i>Subflow_Fwd_Packets</i>
50	<i>FIN_Flag_Count</i>	70	<i>Subflow_Bwd_Packets</i>
51	<i>SYN_Flag_Count</i>	74	<i>Act_data_pkt_forward</i>
52	<i>RST_Flag_Count</i>	-	-

Experiments

● Evaluation Metrics

➤ Certified Robustness

- Mean Certified Radius
- Certified Accuracy

$$\text{Mean Certified Radius (MCR)} = \frac{1}{N} \sum_{i=1}^N R_i$$

$$\text{Certified Accuracy (CerAcc)} = \frac{N_{(F_{\text{smooth}}(x)=y_{\text{true}}) \& (R \geq R_{\text{given}})}}{N}$$

➤ Empirical Robustness

- Robust Accuracy on Adversarial (Malicious) Examples
- Robust Accuracy on Corrupted (Malicious & Benign) Examples

$$\text{Recall} = \frac{TP}{TP + FN}$$

$$\text{Robust Accuracy (RobAcc)} = \frac{N_{(F_{\text{smooth}}(x^*)=y_{\text{true}})}}{N} = \frac{TP + TN}{TP + TN + FP + FN}$$

➤ Regular Predictive Performance

- Clean Accuracy

$$\text{Clean Accuracy (CleAcc)} = \frac{N_{(F_{\text{smooth}}(x)=y_{\text{true}})}}{N}$$

Experiments

● Comparison of Certified Defense Methods

- VRS: Vanilla Randomized Smoothing (ICML 2019) → designed for Image
- FRS: First Order-based Randomized Smoothing (NeurIPS 2020) → designed for Image
- BARS: Boundary-Adaptive Randomized Smoothing (NDSS 2023) → designed for Traffic

Method	Heterogeneity	Universality	Robustness Guarantee Diversity			Adversarial Attacks			Natural Corruptions	
			l_2 Radius	l_1 Radius	l_∞ Radius	l_2 Attack	l_1 Attack	l_∞ Attack	Latency	Loss
VRS [17]	○	●	●	○	○	○	○	○	○	○
FRS [35]	○	●	●	●	●	○	○	○	○	○
BARS [18]	●	●	●	○	○	○	○	●	○	○
MARS	●	●	●	●	●	●	●	●	●	●

[17] Jeremy Cohen, Elan Rosenfeld, and Zico Kolter. 2019. Certified adversarial robustness via randomized smoothing. In International Conference on Machine Learning (ICML). 1310–1320.

[35] Jeet Mohapatra, Ching-Yun Ko, Tsui-WeiWeng, Pin-Yu Chen, Sijia Liu, and Luca Daniel. 2020. Higher-order certification for randomized smoothing. In Advances in Neural Information Processing Systems (NeurIPS). 4501–4511.

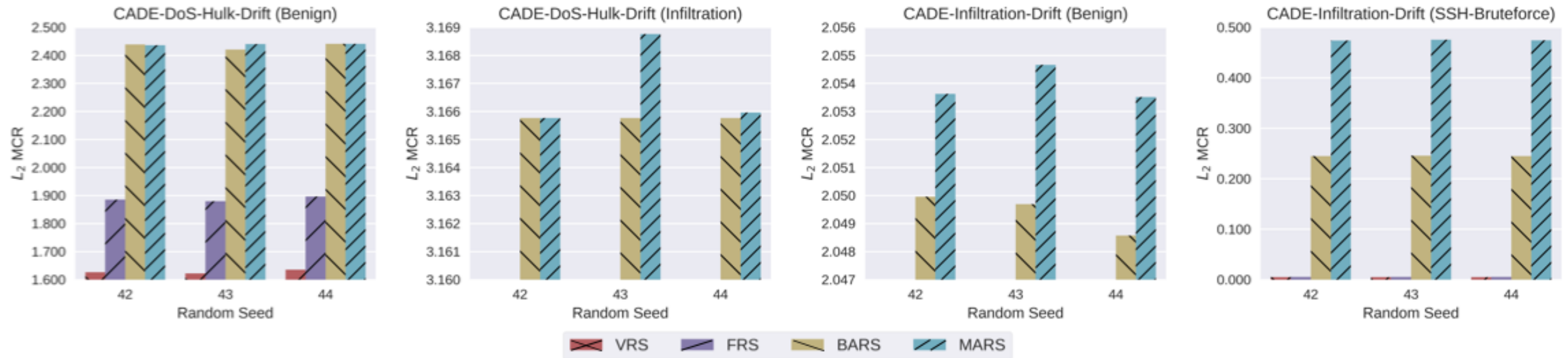
[18] Kai Wang, Zhiliang Wang, Dongqi Han, Wenqi Chen, Jiahai Yang, Xingang Shi, and Xia Yin. 2023. BARS: Local Robustness Certification for Deep Learning based Traffic Analysis Systems. In Network and Distributed Systems Security (NDSS) Symposium.

Evaluation Results and Analysis

$$\text{Mean Certified Radius (MCR)} = \frac{1}{N} \sum_{i=1}^N R_i$$

● Exp 1: Comparison of l_2 -bounded Certified Robustness with SOTA Method

- Exp Setup: $n_{small} = 100$, $n_{large} = 10,000$. Compare the l_2 overall MCR R of the model by category.
- Observation: **MARS always outperforms certified defense baselines VRS, FRS, and BARS.**
 - For CADE trained on DoSHulk-Drift dataset, MARS shows a 0.23% and 0.03% higher MCR in Benign and Infiltration classes, respectively, than SOTA BARS.
 - For CADE trained on Infiltration-Drift dataset, MARS exhibits a 0.22%, 93.66%, and 0.2% MCR increase in Benign, SSH-Bruteforce, and DoS-HULK categories compared to BARS.

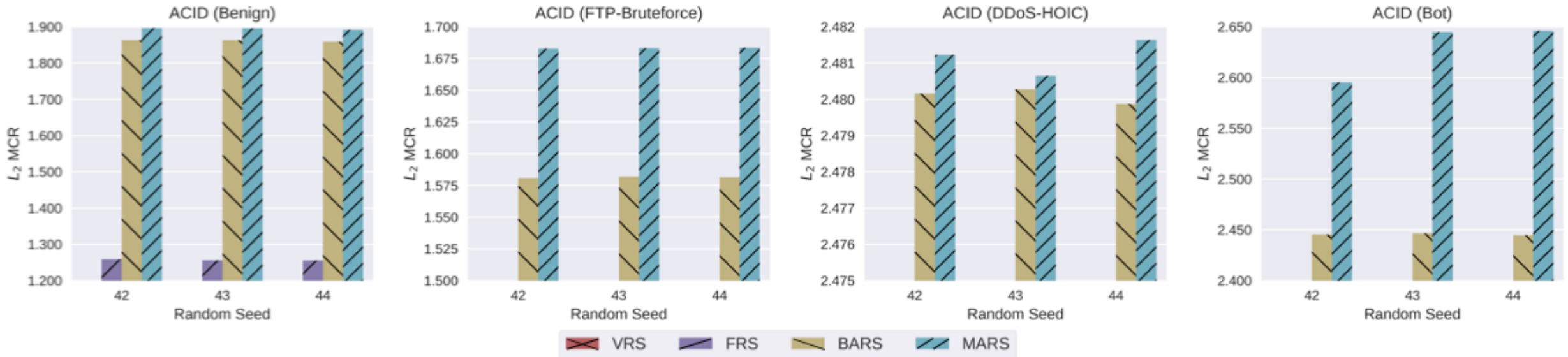


Evaluation Results and Analysis

$$\text{Mean Certified Radius (MCR)} = \frac{1}{N} \sum_{i=1}^N R_i$$

● Exp 1: Comparison of l_2 -bounded Certified Robustness with SOTA Method

- Exp Setup: $n_{small} = 100$, $n_{large} = 10,000$. Compare the l_2 overall MCR R of the model by category.
- Observation: **MARS always outperforms certified defense baselines VRS, FRS, and BARS.**
 - For ACID trained on Diverse Intrusion dataset, MARS exhibits a 1.75%, 6.44%, 0.04%, and 7.49% MCR increase in Benign, FTP-Bruteforce, DDoS-HOIC, and Bot categories compared to SOTA Certified Defense BARS.

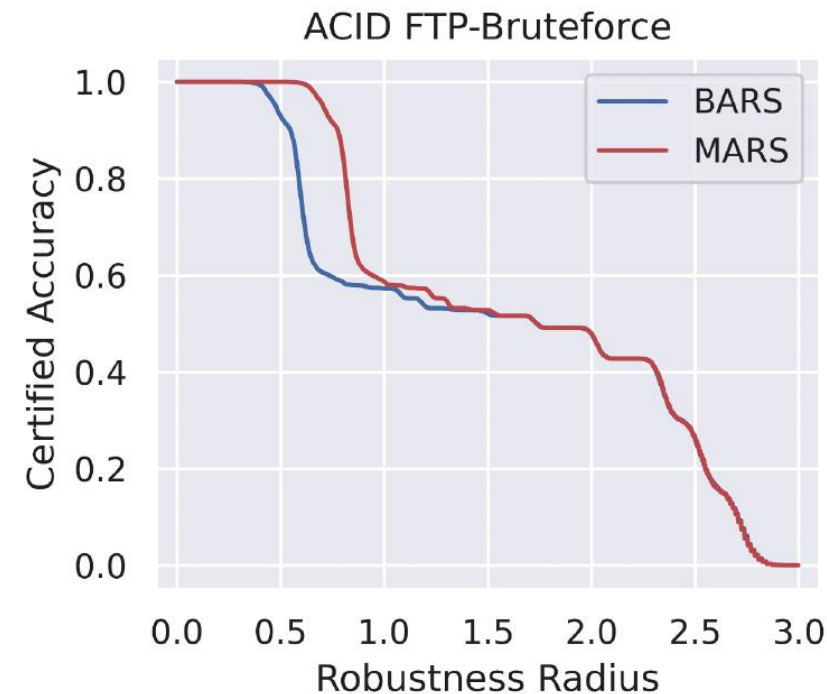
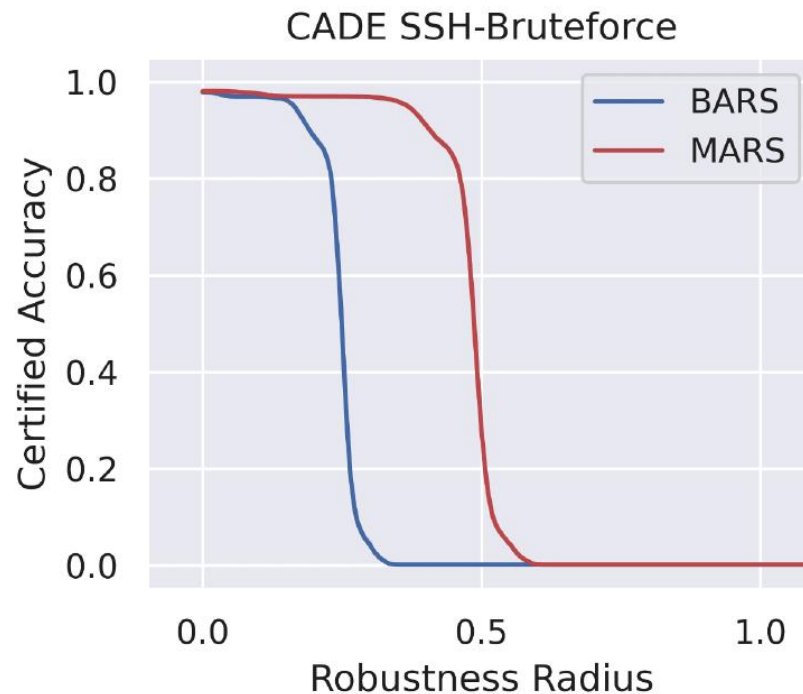


Evaluation Results and Analysis

$$\text{Certified Accuracy (CerAcc)} = \frac{N_{(F_{\text{smooth}}(x)=y_{\text{true}}) \& (R \geq R_{\text{given}})}}{N}$$

● Exp 1: Comparison of l_2 -bounded Certified Robustness with SOTA Method

- Exp Setup: Compare the Certified Accuracy of the model w.r.t the l_2 -bounded certified radius.
- Observation: ***MARS demonstrated the certified robustness of the model in a larger region.***
 - For CADE, MARS maintains 100% accuracy until the MCR threshold reaches 0.4, while the that of the SOTA methods begins to drop sharply when the threshold just exceeds 0.15.
 - For ACID, MARS shows significant advantages over SOTA until the MCR reaches 1.5.

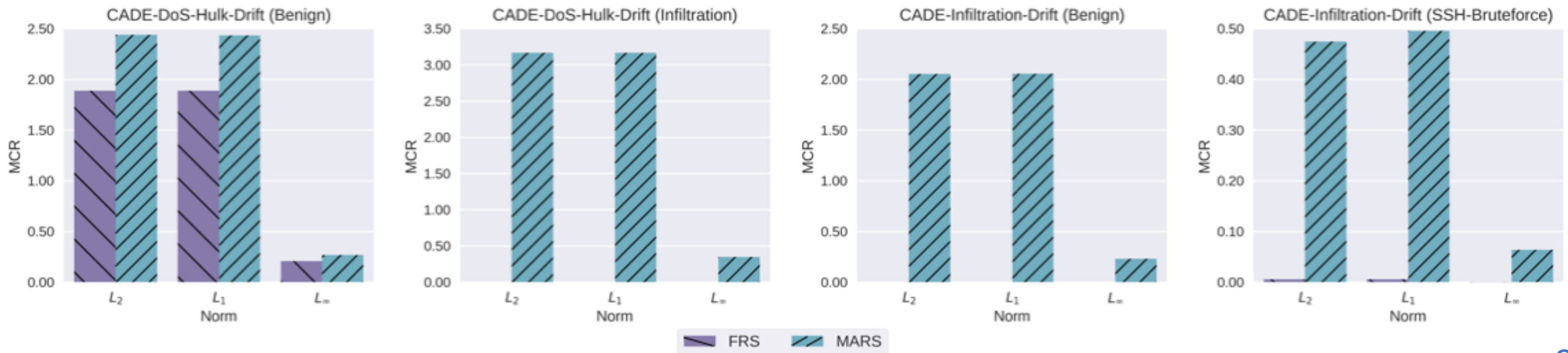


Evaluation Results and Analysis

$$\text{Mean Certified Radius (MCR)} = \frac{1}{N} \sum_{i=1}^N R_i$$

● Exp 2: Comparison of Various l_p -bounded Certified Robustness with SOTA Method

- Exp Setup: $n_{small} = 100$, $n_{large} = 10,000$. Compare the l_1 , l_∞ MCR of the model by category with FRS, since neither VRS nor BARS supports l_1 -bounded and l_∞ -bounded robustness certification.
- Observation: ***MARS consistently provides larger l_p -bounded radius compared to FRS.***
 - FRS fails certification on many classes (MCR=0) due to indiscriminate smoothing of network traffic features, MARS produces non-trivial l_2 , l_1 , and l_∞ radii.
 - For CADE trained on DoSHulk-Drift dataset, MARS outperforms FRS by 29.25%, 28.95%, and 28.72% in l_2 , l_1 , and l_∞ radii on Benign, respectively.

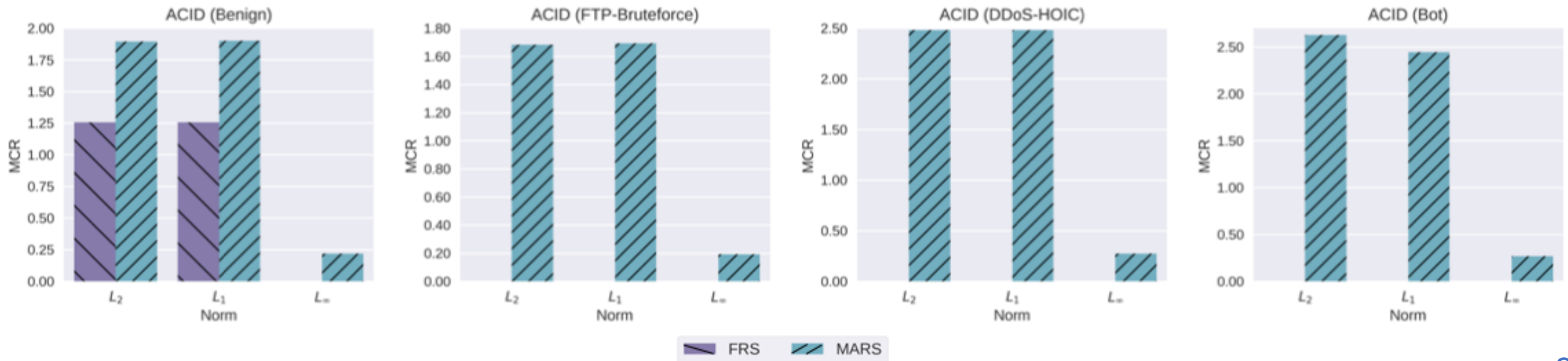


Evaluation Results and Analysis

$$\text{Mean Certified Radius (MCR)} = \frac{1}{N} \sum_{i=1}^N R_i$$

● Exp 2: Comparison of Various l_p -bounded Certified Robustness with SOTA Method

- Exp Setup: $n_{small} = 100$, $n_{large} = 10,000$. Compare the l_1 , l_∞ MCR of the model by category with FRS, since neither VRS nor BARS supports l_1 -bounded and l_∞ -bounded robustness certification.
- Observation: **MARS consistently provides larger l_p -bounded radius compared to FRS.**
 - FRS fails certification on many classes (MCR=0) due to indiscriminate smoothing of network traffic features, MARS produces non-trivial l_2 , l_1 , and l_∞ radii.
 - For ACID trained on Diverse Intrusion dataset, MARS outperforms FRS by 50.78% and 51.32% in l_2 and l_1 radii on Benign, respectively.

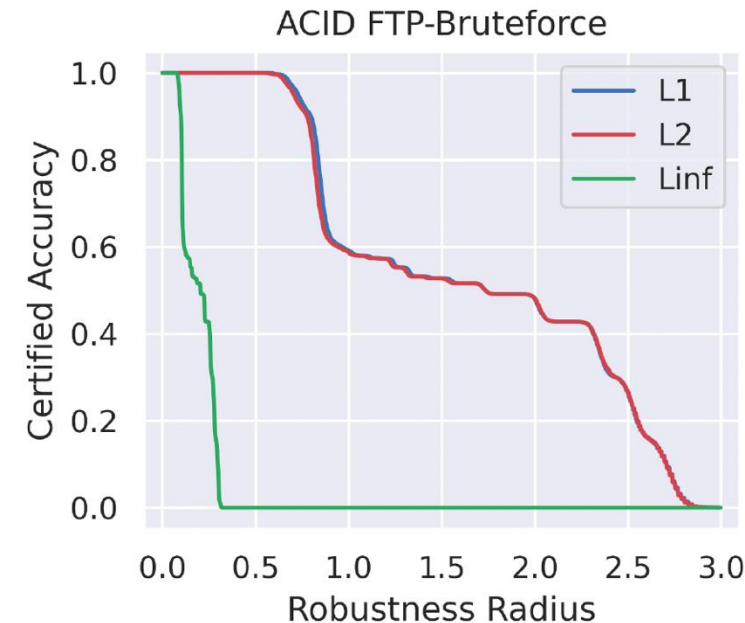
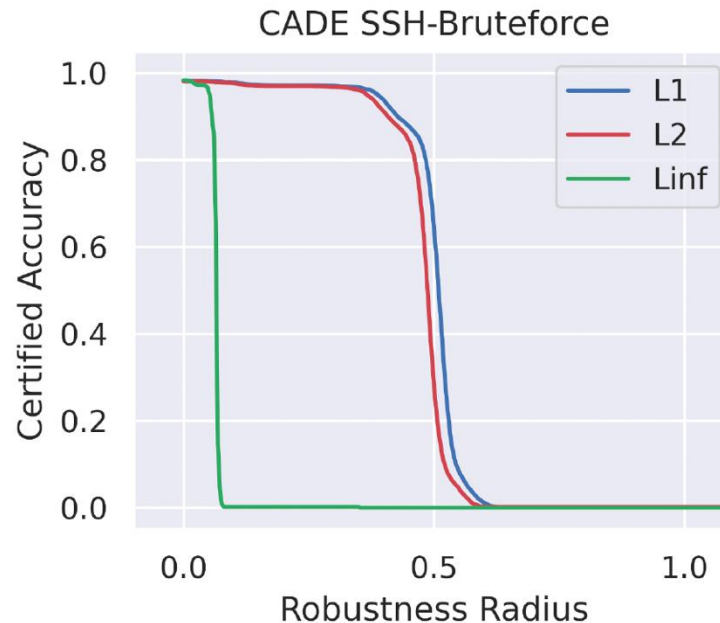


Evaluation Results and Analysis

$$\text{Certified Accuracy (CerAcc)} = \frac{N_{(F_{smooth}(x)=y_{true}) \& (R \geq R_{given})}}{N}$$

● Exp 2: Comparison of Various l_p -bounded Certified Robustness with SOTA Method

- Exp Setup: Compare the Certified Accuracy of the model w.r.t the l_p -bounded certified radius.
- Observation: **l_2 radius is usually smaller than the l_1 radius and larger than the l_∞ radius.**
 - At the same radius, the area bounded by l_1 norm should be the smallest, and the area defined by l_∞ should be the largest.
 - Different norm-bounded radii calculated experimentally are consistent with theoretical results.



Evaluation Results and Analysis

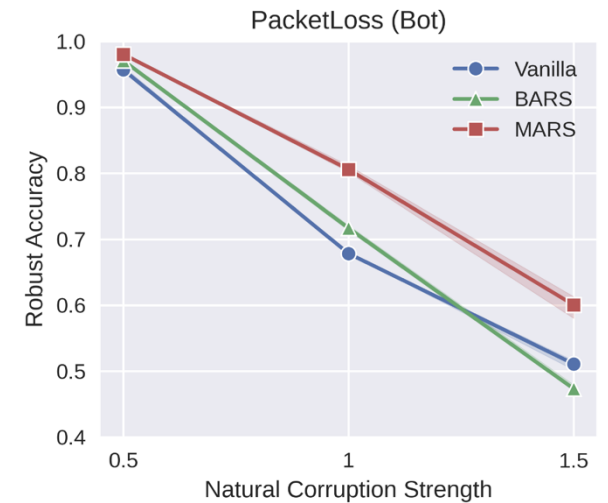
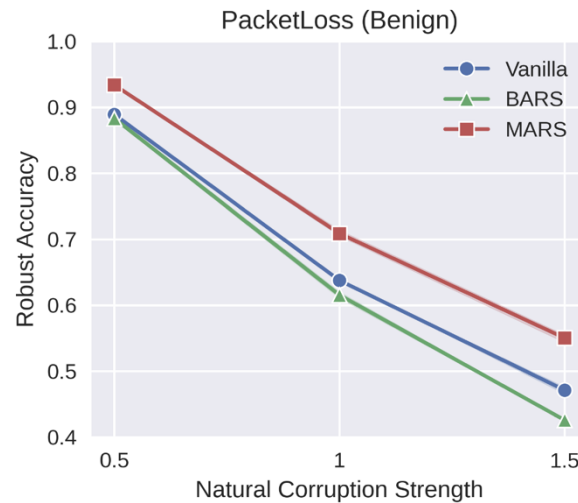
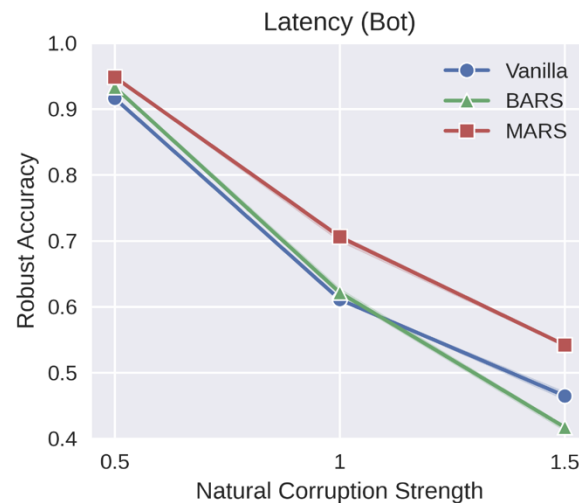
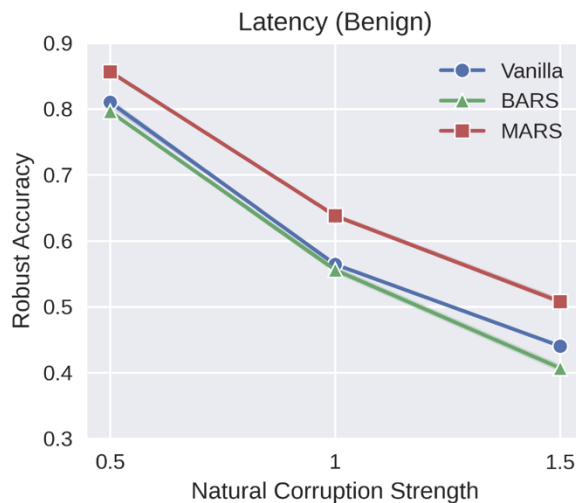
● Exp 3: Comparison of Empirical Robustness against Evasion Attacks with SOTA Method

- Exp Setup: Attack ACID with PGD and EAD adversarial Bot. Iteration is 20. For l_2 -PGD and l_1 -EAD, perturbation limit ϵ is 1.0, with per-step budget ϵ_s of 0.75. For l_∞ -PGD, ϵ is 0.2 and ϵ_s is 0.1.
- Observation: ***MARS surpasses SOTA defense in robustness against evasion attacks.***
 - MARS improves robust accuracy over the Vanilla detector (base model without defense) by 13.79% for l_2 -PGD, 33.94% for l_∞ -PGD, and 10.01% for l_1 -EAD.
 - MARS also outperforms SOTA BARS, boosting robust accuracy by 1.7% for l_2 -PGD, 7.17% for l_∞ -PGD, and 10.11% for l_1 -EAD.
 - MARS well retain the clean accuracy of the ACID on clean Bot samples, reaching 100%.

Method	CleanAcc/Recall on Clean Bot (%)	RobustAcc/Recall on Adversarial Bot (%)		
		l_2 -PGD	l_∞ -PGD	l_1 -EAD
Vanilla	100.00±00.00	83.95±00.00	55.02±00.01	<u>00.27±00.00</u>
BARS [18]	100.00±00.00	<u>96.04±00.05</u>	<u>81.78±00.20</u>	00.16±00.01
MARS	100.00±00.00	97.74±00.13	88.95±00.31	10.28±00.06

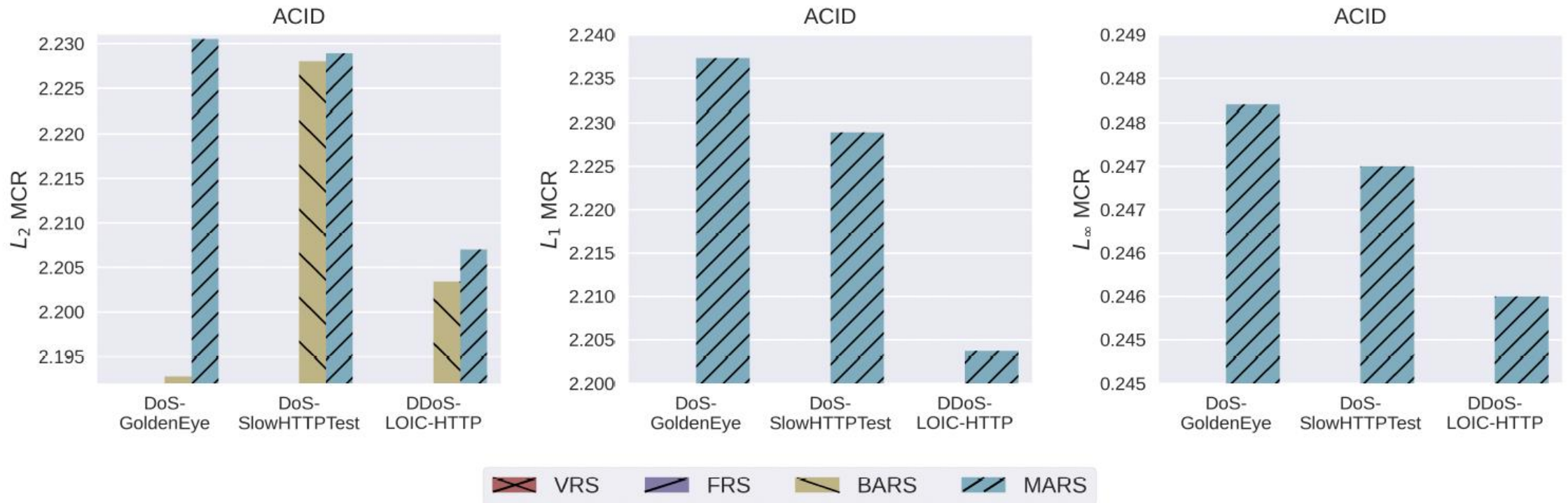
Evaluation Results and Analysis

- **Exp 4: Comparison of Empirical Robustness against Natural Corruptions with SOTA Method**
 - Exp Setup: Generate natural corrupted samples from clean benign/malicious samples using Latency and PacketLoss. Use random noise following a Gaussian distribution with mean 0. Adjust the standard deviation σ in $\{0.5, 1.0, 1.5\}$ to mimic the different corruption strengths.
 - Observation: ***MARS surpasses SOTA in robustness against various corruption intensities.***
 - MARS outperforms SOTA BARS in robust accuracy, exceeding it by 8.53% on corrupted Benign and 7.5% on corrupted Bot.



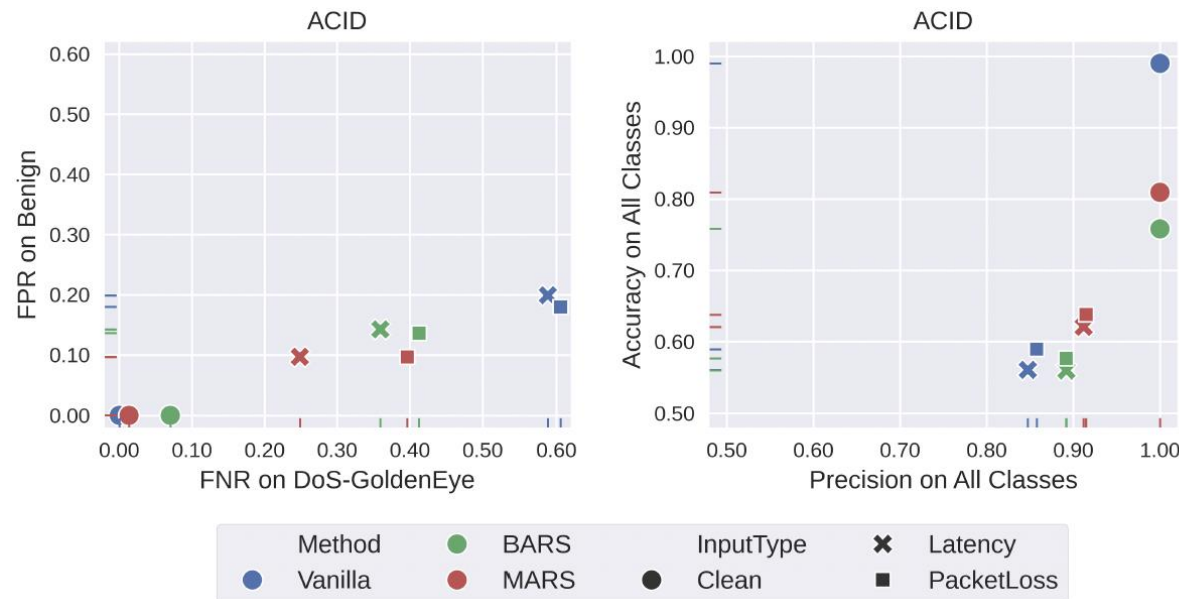
Evaluation Results and Analysis

- **Exp 5: Comparison of l_p -bounded Certified Robustness in Fine-grained Detection with SOTA Method**
 - Exp Setup: Use various DoS-related attacks to extend performance assessment in fine-grained NIDS scenarios. trained a four-class ACID model with 99% accuracy, 100% precision, and an F1-score of 1 on the clean test set. compared l_2 , l_1 , and l_∞ radius.
 - Observation: ***MARS achieved the largest certified radii in all three similar intrusions (DoS-GoldenEye, DoS-SlowHTTPTest, and DDoS-attacks-LOIC-HTTP).***



Evaluation Results and Analysis

- **Exp 6: Comparison of Empirical Robustness against Natural Corruptions in Fine-grained Detection**
 - Exp Setup: Generate natural corrupted samples from clean benign/malicious samples using Latency and PacketLoss. Use random noise following a Gaussian distribution with mean 0, standard deviation 1 to mimic the different corruption strengths.
 - Observation: ***MARS outperforms BARS in defending the classifier against Latency and PacketLoss corruption.*** Accuracy is improved by 6.12% under Latency and PacketLoss corruptions. False Positive Rate (FPR) on corrupted Benign samples decreased by 4.27%, and False Negative Rate (FNR) on corrupted GoldenEye decreased by 6.35%.



Evaluation Results and Analysis

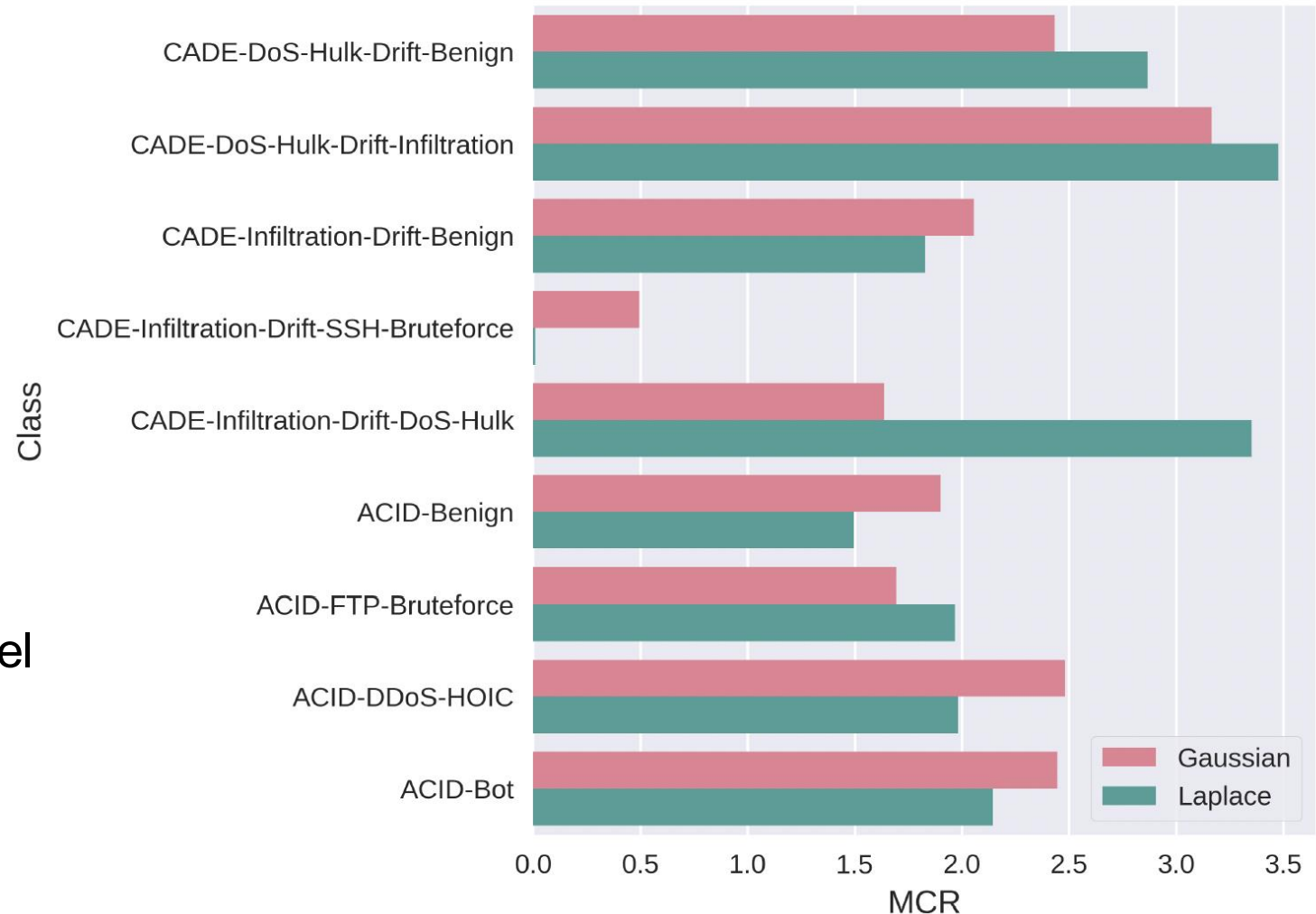
● Exp 7: l_p Certified Robustness with Different Smoothing Distributions

➤ Exp Setup:

- All baselines use Gaussian as the smoothing distribution.
- MARS considers distribution diversity and sequentially uses Gaussian, Laplacian, and Uniform distributions.

➤ Observation:

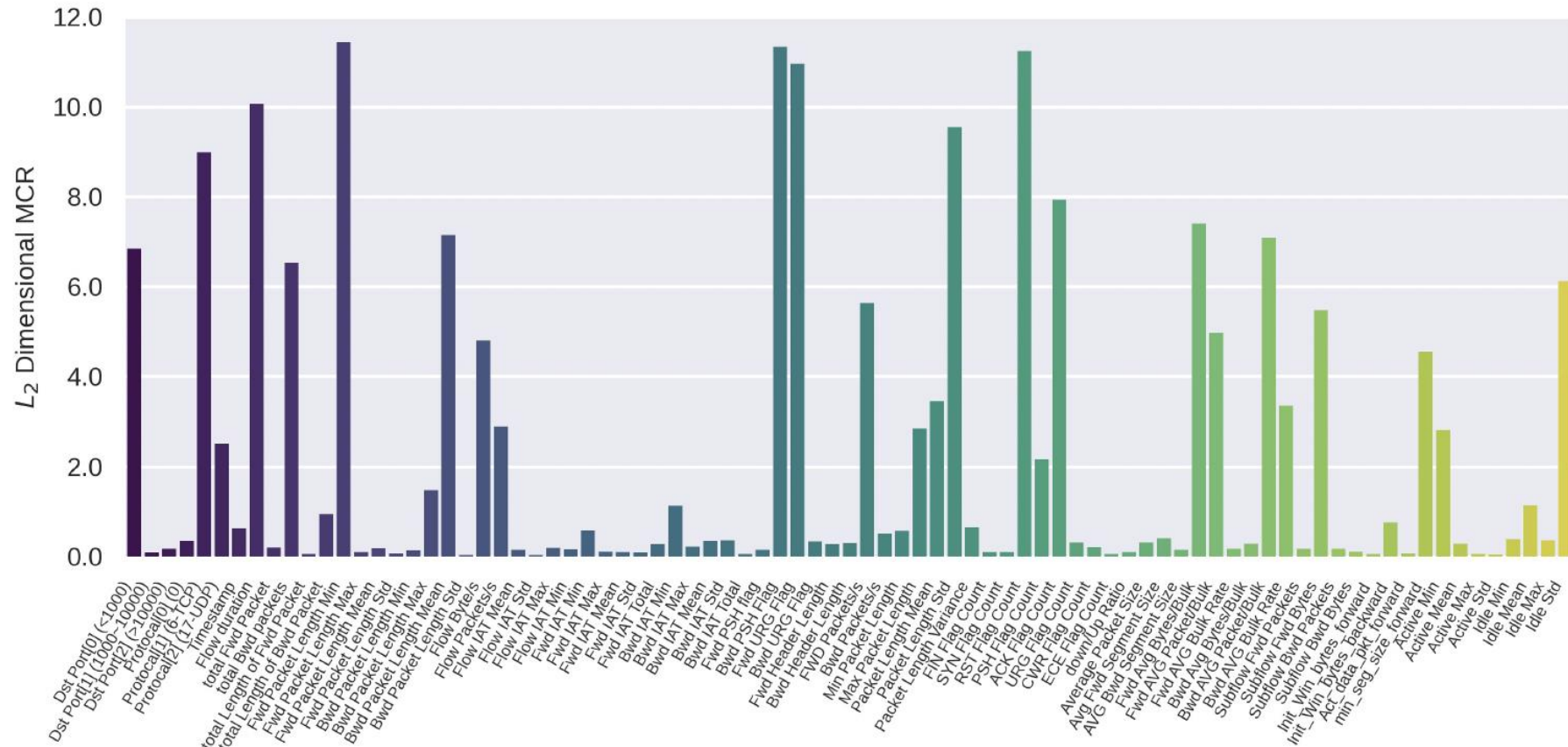
- Different distributions each excel in different classes.
- Using a single distribution may miss a tighter certified radius.



Evaluation Results and Analysis

● Exp 8: Dimension-Wise Certified Robustness

- Exp Setup: obtain l_2 -bounded dimensional radius R_i of the ACID on the GoldenEye sample.
- Observation: A smaller dimensional radius indicates greater sensitivity and importance of the input feature dimension to the NIDS, as such features are more likely to alter the predictions.



Evaluation Results and Analysis

● Exp 8: Dimension-Wise Certified Robustness

- Exp Setup: MARS's Top-5 and bottom-5 dimension-wise radius of the ACID.
- Observation:
 - The model demonstrates greater sensitivity to *inter arrival time (IAT)*-related features while showing greater robustness to *forward packet length-related* features.
 - This finding is consistent with the previous observation that the vanilla ACID model exhibited significantly reduced robust accuracy on corrupted samples using Latency.

No	Radius	FeatureName	Description
24	0.0426	Flow_IAT_Std	Standard deviation time two flows.
20	0.0433	Bwd_Packet_Length_Std	Standard deviation size of packet in backward direction.
79	0.0488	Active_Std	Standard deviation time a flow was active before becoming idle.
72	0.0569	Init_Win_bytes_forward	Number of bytes sent in initial window in the forward direction.
78	0.0576	Active_Max	Maximum time a flow was active before becoming idle.
8	10.0741	Flow_Duration	Flow duration.
39	10.9644	Fwd_URG_Flag	Number of times URG flag was set in packets travelling in the forward direction (0 for UDP).
52	11.2367	RST_Flag_Count	Number of packets with RST.
38	11.3300	Bwd_PSH_Flag	Number of times PSH flag was set in packets travelling in the backward direction (0 for UDP).
13	11.4358	Fwd_Packet_Length_Min	Minimum size of packet in forward direction.
All	2.2305	MCR	Mean certified radius per class.

Evaluation Results and Analysis

● Exp 9: Certification Time Overhead

- Exp Setup: Evaluate the Average Certification Time (ACT/seconds) per sample and mean certified radius (MCR) of network intrusion models (CADE and ACID) on different types of network traffic.
- Observation: **MARS achieves the highest MCR across all competitions, with an average increase in ACT per sample of 21.1 milliseconds (ms).** For CADE, ACT per sample ranges from 8.0 to 14.6 ms, while for ACID, ACT per sample ranges from 24.4 to 36.1 ms. MARS-defend CADE takes approximately 25 seconds to certify 2K Infiltration or SSH-Bruteforce samples and around 125 sec for 13K benign or 10K DoS-Hulk samples. For MARS-defend ACID, the certification times are about 300 sec for 13K benign/DDoS-HOIC or 11K DDoS-LOIC-HTTP samples, 200 sec for 6K Bot or 7K DoS-GoldenEye samples, and 100 sec for 3K FTP-Bruteforce or DoS-SlowHTTPTest samples

Table 5. l_2 -bounded MCR and ACT/seconds per sample of CADE.

MCR (ACT)	Benign	SSH-Bruteforce	Infiltration	DoS-Hulk
VRS	0.0000 (0.0008)	0.0049 (0.0015)	0.0000 (0.0017)	1.4082 (0.0012)
FRS	0.0000 (0.0089)	0.0054 (0.0125)	0.0000 (0.0122)	1.6316 (0.0134)
BARS	2.0494 (0.0004)	0.2450 (0.0011)	3.1658 (0.0013)	1.6372 (0.0007)
MARS	2.0539 (0.0096)	0.4744 (0.0133)	3.1668 (0.0131)	1.6406 (0.0146)

Table 6. l_2 -bounded MCR and ACT/seconds per sample of ACID.

MCR (ACT)	Benign	FTP-Bruteforce	DDoS-HOIC	Bot	DoS-GoldenEye	DoS-SlowHTTPTest	DDoS-LOIC-HTTP
VRS	0.1950 (0.0028)	0.0000 (0.0045)	0.0000 (0.0028)	0.0000 (0.0040)	0.0001 (0.0039)	0.0000 (0.0053)	0.0000 (0.0033)
FRS	1.2077 (0.0238)	0.0000 (0.0323)	0.0000 (0.0235)	0.0000 (0.0565)	0.0024 (0.0309)	0.0014 (0.0459)	0.0000 (0.0269)
BARS	1.9036 (0.0029)	1.5814 (0.0046)	2.4801 (0.0029)	2.4456 (0.0041)	2.1928 (0.0040)	2.2280 (0.0051)	2.2034 (0.0033)
MARS	1.9260 (0.0253)	1.6832 (0.0346)	2.4812 (0.0244)	2.6288 (0.0327)	2.2305 (0.0331)	2.2289 (0.0361)	2.2070 (0.0279)

Discussion

- Applications of the Dimensional Certified Radius

- The certified radius reflects the robustness of DNNs in NIDS against potential input traffic variations, which is crucial for reliable network intrusion detection.
- A larger radius indicates stronger robustness, aiding prompt responses to complex intrusions, while a smaller radius highlights vulnerabilities, guiding targeted defenses.
- The certified dimensional radius pinpoints vulnerabilities in specific features, enabling focused defenses and reducing learning overhead on robust patterns.

- Importance of the Norm Diversity

- Calculating the certified radius under various norms (l_2 , l_1 , and l_∞) provides a comprehensive evaluation of robustness against different attack vectors.
- The l_2 radius reflects robustness to overall feature variation, l_1 to sparse perturbations, and l_∞ to extreme outliers.
- In an environment with evolving cyber threats, multi-norm certification is valuable for preparing for dynamically changing attacks.

Conclusion

● Contribution

- Multi-Order Information Utilization
 - We design an adaptive randomized smoothing algorithm based on zero-order and first-order information of the smoothed classifier to calculate robust radii, yielding tighter lower bounds of robustness compared to existing methods.
- Dimensional-Wise Robust Radius
 - We propose a dimension-wise robust radius calculation algorithm based on the sensitivity of each feature dimension, enabling fine-grained robustness certification for heterogeneous input features.
- New Threat Model
 - Extended empirical robustness evaluation of traffic classifier to account for natural corruption (e.g., Latency and Packet Loss) in addition to evasion attacks using adversarial examples.
- Extensive Experiments
 - Experiments on various network intrusion detectors show that the proposed method improving certified robustness against diverse adversarial attacks and natural corruption.

Thank You!

Mengdie Huang (Maggie)
huan1932@purdue.edu